

Problem

Let a, b, c, d be pairwise distinct positive integers such that

$$\frac{a}{a+b} + \frac{b}{b+c} + \frac{c}{c+d} + \frac{d}{d+a}$$

is an integer.

Prove that $a + b + c + d$ is not a prime number.

Source: Original Problem by Aryan Goutam

§1 Constructing a Complementary Sum

Define

$$X = \frac{a}{a+b} + \frac{b}{b+c} + \frac{c}{c+d} + \frac{d}{d+a}.$$

By hypothesis, X is an integer.

Now introduce the complementary quantity

$$Y = \frac{b}{a+b} + \frac{c}{b+c} + \frac{d}{c+d} + \frac{a}{d+a}.$$

Since each pair of fractions adds to 1, we obtain

$$X + Y = 4.$$

Hence Y is also an integer.

§2 Determining the Value of the Sum

Since all variables are positive,

$$\frac{a}{a+b} > \frac{a}{a+b+c+d},$$

and similarly for the remaining three fractions. Therefore

$$X > \frac{a+b+c+d}{a+b+c+d} = 1.$$

Likewise,

$$Y > 1.$$

Since X and Y are integers satisfying

$$X + Y = 4,$$

the only possibility is

$$X = Y = 2.$$

Consequently,

$$X - Y = 0.$$

§3 Extracting the Hidden Relation

We compute

$$0 = \frac{a-b}{a+b} + \frac{b-c}{b+c} + \frac{c-d}{c+d} + \frac{d-a}{d+a}.$$

Group the first and third fractions together and the second and fourth together:

$$0 = \left(\frac{a-b}{a+b} + \frac{c-d}{c+d} \right) + \left(\frac{b-c}{b+c} + \frac{d-a}{d+a} \right).$$

A direct computation yields

$$\frac{a-b}{a+b} + \frac{c-d}{c+d} = \frac{2(ac-bd)}{(a+b)(c+d)},$$

and

$$\frac{b-c}{b+c} + \frac{d-a}{d+a} = \frac{2(bd-ac)}{(b+c)(d+a)}.$$

Hence

$$\begin{aligned} 0 &= 2(ac-bd) \left(\frac{1}{(a+b)(c+d)} - \frac{1}{(b+c)(d+a)} \right) \\ &= 2(ac-bd) \cdot \frac{(b+c)(d+a) - (a+b)(c+d)}{(a+b)(c+d)(b+c)(d+a)}. \end{aligned}$$

The numerator simplifies to

$$(b+c)(d+a) - (a+b)(c+d) = (a-c)(b-d).$$

Therefore

$$0 = 2(ac - bd) \cdot \frac{(a-c)(b-d)}{(a+b)(b+c)(c+d)(d+a)}.$$

Since the numbers a, b, c, d are pairwise distinct,

$$(a-c)(b-d) \neq 0.$$

The denominator is positive, so we conclude that

$$ac = bd.$$

§4 Applying the Four Number Lemma

Since

$$ac = bd,$$

the Four Number Lemma implies that there exist positive integers p, q, r, s such that

$$a = pq, \quad b = pr, \quad c = rs, \quad d = qs.$$

Substituting into the sum gives

$$\begin{aligned} a + b + c + d &= pq + pr + rs + qs \\ &= p(q + r) + s(q + r) \\ &= (p + s)(q + r). \end{aligned}$$

Because a, b, c, d are pairwise distinct positive integers, we have

$$p + s > 1, \quad q + r > 1.$$

Thus $a + b + c + d$ is expressed as a product of two integers greater than 1.

Hence

$$a + b + c + d$$

is composite.

§5 The Final Conclusion

Therefore,

$a + b + c + d$ is not a prime number.